H.D.R.A Report

Hazards and Disasters Risk Assessment

1. Financial Impact (Present Value Capitalization)

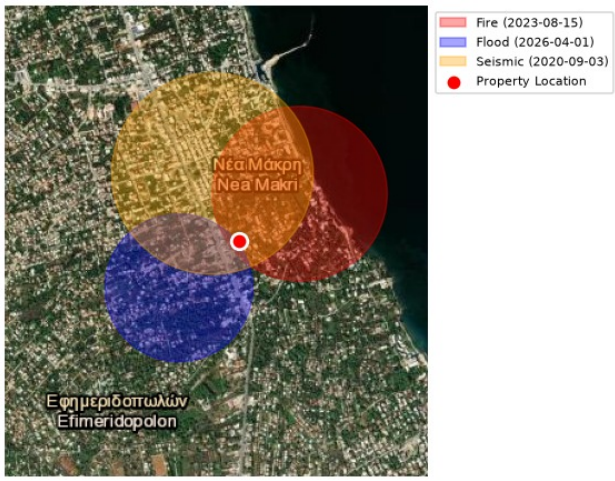
SLE: 2325.0 EUR | ARO (50yr ref): 0.2468 | ALE: 573.75 EUR/year

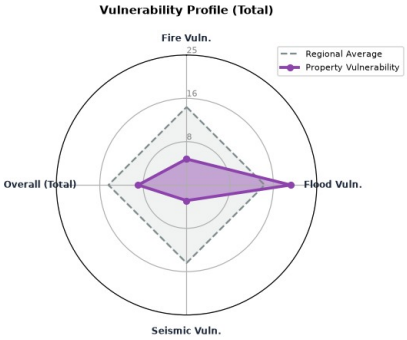
Final Property Value: 21550.0 EUR (Immediate Market Drop: -13.8%)

** SLE (Single Loss Expectancy): Expected financial loss from a single disaster event.*
** ARO (Annualized Rate of Occurrence): Estimated annual event frequency based on 50-year historical data.*
** ALE (Annualized Loss Expectancy): Expected annual financial cost of risk (SLE x ARO).*

2. Details

Attribute	Value
Scenario	Total
Initial Property Value	25000.0 EUR
Floor Level	Ground
Slope (EU-DEM 25m)	3.4%
Construction Material	Concrete
Construction Standards	Post-1999





3. Structural Impact Factors (Hierarchical Weighting)

Property Characteristic	Calculated Impact / Vulnerability Penalty
Material: Concrete	High structural resilience (+0% penalty)
Standards: Post-1999	Modern anti-seismic code (-2% Seismic)
Floor: Ground	Moderate Flood vulnerability (+15% Flood)
Slope: 3.4%	Optimal drainage & stability (+0% penalty)
Roof: Tiles	Minor ember penetration risk (+3% Fire)
Vents: None	Sealed against fire embers (-3% Fire)

4. Last Recorded Disaster Metadata (8-Mile Buffer / 50-Year Horizon)

--- Fire Hazard ---
Data Source: Hellenic Fire Service / Copernicus EFFIS (Retrieved: 2026-10-14) | Event Date: 2023-08-15 | Total Events (50yrs): 12
Event Details: Burned area: 4,500 hectares within 8-mile buffer. High wind velocity.

--- Flood Hazard ---
Data Source: Copernicus ERA5 / Ministry of Climate Crisis (Live API) (Retrieved: 2026-10-14) | Event Date: 2026-04-01 | Total Events (50yrs): 18
Event Details: Rainfall: 58.5mm in 24h (Peak: 65.4mm). Severe flash flood risk.

--- Seismic Hazard ---
Data Source: EMSC / Geodynamic Institute of Athens (Live API) (Retrieved: 2026-10-14) | Event Date: 2020-09-03 | Total Events (50yrs): 13
Event Details: Magnitude: 4.3 Mw. Focal depth: 7.0km.

Terms & Privacy (GDPR): This H.D.R.A. report is an automated econometric & geospatial estimation based on Copernicus/EMSC/EFFIS data. It does not substitute a certified property appraisal or structural engineering inspection. Coordinates are processed anonymously.